Dark web threat actor de-anonymization

Industry Context

The dark web has become a significant operating environment for cybercriminals and threat actors because technologies such as Tor hidden services help conceal their identities. These actors may engage in illegal activities such as selling stolen data, providing hacking services, financial fraud, and money laundering. Identifying them and linking their online activities to real-world entities remains a major challenge for cybersecurity investigators.

Threat actors often use multiple usernames, cryptocurrency wallets, PGP keys, and online marketplaces to hide their identities. They may also change their personas or migrate between platforms, making it difficult to track their activities. These challenges can lead to:

- Difficulty identifying threat actors across multiple platforms
- Fragmented intelligence and time-consuming investigations
- Challenges linking different online identities
- Delays in detecting suspicious relationships
- Difficulty establishing reliable attribution

Current investigations often rely on manual data collection and analysis, which can be slow and inefficient. There is a growing need for an intelligent platform that automatically collects and analyses threat intelligence, identifies potential relationships between online personas, and presents findings through dashboards and relationship graphs. By using AI-assisted analysis and timeline-based searches, the proposed system aims to help authorized investigators **identify suspicious connections, prioritize investigative leads, and generate intelligence reports** while maintaining evidence traceability and human verification.

Problem Statement

The dark web provides anonymity to users through technologies such as Tor hidden services. This makes it difficult for cybersecurity investigators to identify and connect threat actors involved in illegal activities.

The proposed system aims to collect information about threat actors from multiple sources such as dark-web marketplaces, forums, and other publicly available intelligence sources. The collected information can include usernames or handles, PGP keys, cryptocurrency wallet identifiers, infrastructure indicators, and relationships between different online identities.

The system will analyze these indicators and create connections between different identities. It will also use AI-based techniques such as **stylometric analysis and behavioural profiling** to identify possible connections between personas that may belong to the same threat actor.The system is intended to support authorized cybersecurity investigations by providing an analytical platform rather than relying on a single indicator.

Objectives

The main objectives are:

1. Collect threat-actor intelligence from multiple reliable sources.
2. Maintain information about handles, PGP keys, wallets and other identifiers.
3. Identify relationships between different online personas.
4. Detect infrastructure-related indicators associated with hidden services.
5. Apply AI-based analysis to identify possible persona relationships.
6. Assign an attribution confidence level to identified relationships.
7. Provide a searchable relationship graph and analytical dashboard.
8. Allow investigators to analyze information across a selected timeline.
9. Support autonomous and continuous intelligence collection.
10. Export investigation results in **CSV, JSON and report formats**.

Expected Outcomes

1. Develop an end-to-end Dark Web Threat Intelligence System for collecting, storing, and analyzing threat actor information.
2. Provide a GUI/dashboard for searching, querying, and visualizing threat intelligence.
3. Maintain information such as actor profiles, handles, PGP keys, wallets, infrastructure indicators, persona linkages, attribution confidence, category, scan date, and sources.
4. Enable contextualization and correlation of threat actor data for better analysis.
5. Provide secure and efficient data storage, retrieval, and management.
6. Allow users to export results in CSV, JSON, and report formats.
7. Apply key software engineering principles such as security, scalability, maintainability, reliability, and usability.
8. Demonstrate proper requirements analysis, system design, SDLC selection, and evaluation of risks, limitations, and future enhancements.